

poor maintenance and, in many situations, un-upgraded power systems to meet high annual demand and growth rates. Investments in the IIoT for both existing and new equipment can significantly reduce unscheduled downtime of the grid, thereby improve reliability and reduce costs.

In conventional power plants, the IIoT can be used to balance production with the life of equipment and lifecycle cost of maintenance.

GE's gas and coal power plant captures the unique characteristics of the IIoT. Its digital technologies, when applied to the new coal and gas fired power plant, have increased fuel efficiencies by three per cent and power output by two per cent.

In a non-conventional wind power plant, the IIoT can be used for reducing wake losses by adjusting pitch and yaw angles of individual turbines, controlling stress and fatigue loading within acceptable limits, and setting individual turbines in wind farms to optimum level-to-local conditions.

The IIoT orientation across oil and gas industries

Asset-intensive industries like oil and gas, mining, construction, energy, rail and logistics face many challenges. These must be solved to improve reliability, operational performance and achieve operational objectives.

In the oil and gas industry, asset monitoring is critical, during oil exploration to ensure expensive rigs are in working order. Parts and corrective maintenance, and timely oil pumping is necessary too. Equipment used should meet various health and environmental regulations. As a result, gathering of data is extremely essential during transmission from exploration to production sites. In general, an oil and gas company must record all details of its assets and all activities of those assets to maintain and maximise site productivity and asset performance.

As a part of IIoT initiatives, in Eagle Ford region of North America, a new suite of technologies associated with the IoT enabled oil rigs to produce 65 times more efficiently in 2013.

Despite margin pressure and capital constraints, the oil and gas industry

has many reasons to invest in this front-end IoT technology. Gathering of geo-technical seismic related data for reserves/deposits of oil and gas need digital technology. The IIoT, which is a new approach for Big Data and predictive analytics, certainly is a boon here. Compressors, turbines and other gas transmission equipment are becoming smarter due to the application of IIoT technologies.

As far as safety and regulatory issues are concerned, IoT solutions can greatly manage remote facilities and tank collection sites in real-time frameworks.

In oil-extraction facilities and tank collection sites, oil and gas companies must manage spills, emergency shut downs and regulatory issues. Most of these are remote field operations. Today, sensors can help with predictive maintenance, by reaching problem areas and gathering information about pressure and flow rates. Once problems are identified, corrective actions can be taken.

A prominent problem facing gas wells is liquid loading, which is the inability of gas to remove liquids being produced in the well bore. A hole is drilled for the purpose of exploration. This problem occurs when velocity of the gas being produced drops below critical levels. Accumulated liquid at one stage ceases production. Early prediction is necessary in this case. This is where the IIoT comes into play.

The IIoT setting new standards

The main requirements of the industry are capital and demand for product. As such, the IIoT is still a dream for small-scale industries. However, this technology can add new colours to the industry. Companies often face problems like market speed, volatility and disruptive technologies. Those using the IIoT can significantly improve safety, efficiency and profitability of their products.

The IIoT enhances manufacturing and industrial processes. While communicating with real-time data, smart machines are better than humans. Real-time data increases efficiency, solves problems, saves time and money, and supports business intelligence efforts. **EFY**



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